Answer key: Lesson 2, Build the Figure

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the picture of 8 shapes. Count each kind. Fill in the blanks.

Answer: 4 squares; 3 triangles; 1 circle; 1 more square than triangles

There are 4 squares, 3 triangles and 1 circle: 4 + 3 + 1 = 8 shapes. 4 squares is 1 more than 3 triangles.

Watch for: Counting a shape twice or missing one.; Writing the total, 8, in one of the blanks.; Writing 4 for the difference.

S2. A square is cut into 2 triangles. Use your ruler. Draw the line.

Answer: a line from one corner to the opposite corner; half

A line from one corner to the opposite corner cuts the square into 2 triangles that match. Each triangle is 1 of 2 equal parts, so each is a half of the square.

Watch for: Drawing a line across the middle: that makes 2 rectangles.; Writing quarter: there are only 2 parts.

S3. Build the boat with squares and triangles. Use as few triangles as you can. Fill in the blanks.

Answer: 2 triangles; 2 squares

Each slanted end needs 1 triangle, so there are 2 triangles. The middle is 2 squares side by side. The boat is 2 triangles and 2 squares.

Watch for: Using squares for the slanted ends.; Writing 4 triangles (2 more for the middle).

S4. The rectangle is made of squares. Fill in the blanks.

Answer: 4; 3; 12

There are 4 squares in each row and 3 rows: 4, 8, 12. So there are 12 squares in all.

Watch for: Writing 7 (4 + 3).; Counting only one row.

S5. Fill in the blanks. Circle one word each time.

Answer: rectangle; circle; half circle

2 squares side by side make a rectangle. 2 half circles with their straight sides together make a circle. 2 quarter circles with their straight sides together make a half circle.

Watch for: Writing square for 2 squares.; Writing circle for 2 quarter circles: it takes 4.

Think harder (page 3)

T1. Use your ruler. Split the figure into its shapes.

Answer: a line across at the bottom of the roof; 2 more lines through the middle

Draw a line across from one side of the house to the other, where the roof ends. That splits the figure into 1 triangle and 1 big square. Then draw 1 line down the middle and 1 line across the middle of the big square: that makes 4 small squares.

Watch for: Drawing a line down the middle of the whole house.; Drawing only 1 extra line in the square.

T2. Count each kind of shape in the picture. Fill in the table.

Answer: triangles 5; squares 6; circles 3

Triangles: 3 and 2 make 5. Squares: 2 and 4 make 6. Circles: 1 and 2 make 3. 5 + 6 + 3 = 14 shapes in all.

Watch for: Counting the tiles instead of the shapes.; Counting a tile that has 4 shapes as 1.

T3. Every Triangle. The square is turned on its corner. It has 2 lines from corner to corner. Count the small triangles and the big triangles. Fill in the blank.

Answer: 8

There are 4 small triangles, each with the middle point as one corner. There are 4 big triangles, each made of 2 small ones and using 3 corners of the square: A B C, B C D, C D A and D A B. 4 and 4 make 8 triangles in all.

Watch for: Counting only the 4 small triangles.; Counting a big triangle twice.

Go further (page 4)

G1. The rectangle is 2 rows of 3 squares. Build it with the fewest pieces. Use squares and rectangles from your kit.

Answer: 6; 3

With squares only it takes 6 pieces. A kit rectangle covers 2 squares, so 3 rectangles cover 6 squares. 3 pieces is the fewest, because no piece covers more than 2 squares.

Watch for: Writing 6 for the fewest.; Writing 2 (2 rectangles cover only 4 squares).

G2. The staircase is made of 6 squares. Which sets of pieces make it? Tick True or False for each set.

Answer: true; false; true; false

6 squares make it: True. 3 rectangles cannot make it: after the top square and the bottom right square each take a neighbour, the 2 squares left over do not touch, so they need 2 squares, not 1 rectangle: False. 2 rectangles and 2 squares cover 4 + 2 = 6 squares: True. 2 rectangles and 3 squares cover 7 squares, too many: False.

Watch for: Saying True for any set with the right number of squares but the wrong shape.; Not checking that the total is 6.

G3. Draw a figure for a partner. It has exactly 2 triangles, 1 square and 1 rectangle. Many answers are right.

Answer: any figure with exactly 2 triangles, 1 square and 1 rectangle

A house works: a rectangle for the wall, a square for a window, and 2 triangles for the roof. Count each kind to check your figure: 2 triangles, 1 square, 1 rectangle.

Watch for: Drawing 2 rectangles.; Not counting the shapes at the end.

Investigation (page 5): 2 triangles, many shapes

First: a square (the 2 longest sides touch). Then 3 different shapes in all when whole sides touch: a square, a bigger triangle (2 short sides touch, the right-angle corners together) and a slanted four-sided shape (2 short sides touch, turned the other way). A child who finds 2 has the main idea; a child who finds the third has seen that it depends on which sides touch.

Look for: A child who tries different sides and draws each result, who says the shape depends on which sides touch, and who checks that whole sides touch. Credit any clear reason. Do not mark a child down for stopping at 2 shapes.

Exit ticket (page 5)

E1. Mei makes a picture with 3 squares, 2 triangles and 2 circles. How many shapes are in the picture? Fill in the blank.

Answer: 7

3 squares, 2 triangles and 2 circles: 3 + 2 + 2 = 7 shapes.

Watch for: Writing 3, the squares only.; Writing 5 (leaving out the circles).

E2. Can 2 triangles make a square? Circle Yes or No.

Answer: Yes

Yes: when the 2 longest sides touch, 2 triangles make a square.

Watch for: Saying No because they are not squares.; Not trying it with the pieces.

E3. A rectangle is made of 6 squares. Mei makes it again with rectangles that are each 2 squares long. How many rectangles does she need?

Answer: 3

Each rectangle covers 2 squares. 2, 4, 6: she needs 3 rectangles.

Watch for: Writing 6, the squares.; Writing 2.

Sneak-peek puzzles (not in the workbook)

1. 2
2. A square, or a bigger triangle, or a slanted four-sided shape. Many answers are right: it depends on which sides touch.